PRODUCT PROPOSAL - MVP - HEALTHCARE

AI Healthcare Receptionist

Telegram-Based AI Agent

A focused, approval-led assistant for patient appointments and clinic information management.

TEAM

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Version 1.0 - Simplified project proposal

01 / CONTEXT

Problem Statement

Small clinics often handle appointment questions, doctor discovery and policy clarifications through fragmented manual conversations. The result is delayed replies, repeated staff work and inconsistent information.

Target users

Patients who need clinic information or appointment help.

Clinic managers who maintain doctors, schedules, fees and policies.

Why Telegram

A familiar text channel makes the MVP accessible in Hindi, English and Hinglish without building calling, audio streaming or telephony infrastructure.

Proposed Solution

The **AI Healthcare Receptionist - Telegram-Based AI Agent** is a focused conversational assistant. It answers approved clinic FAQs, helps patients discover doctors and manages verified appointment workflows. Clinic managers use the same channel to maintain approved information and review document-extracted drafts.

Patient assistant	Clinic-management assistant
FAQs, doctors, availability, booking confirmation, lookup, cancellation and rescheduling. Uncertain requests go to staff.	Clinic setup, doctors, schedules, fees, contact details and document review/approval through guided Telegram conversations.

Conversation boundary

The assistant supports administrative healthcare access - not clinical judgement. It never gives diagnoses or prescriptions. Telegram is the **only user-facing channel** in this MVP; patients should share minimal data and protected actions require secure backend verification.

02 / CAPABILITIES

Core Features

The scope is intentionally limited to the highest-value reception and clinic-information workflows.

Patient appointment workflows

Capability	MVP behaviour
Approved clinic FAQs	Answers only from manager-approved clinic information; unknown or sensitive requests are escalated.
Doctor discovery	Shows doctors, specialties, consultation fees and approved clinic details.
Availability and booking	Checks slots, presents options and creates an appointment only after explicit patient confirmation.
Appointment support	Supports verified lookup, cancellation and rescheduling; consequential changes need confirmation.

Clinic-management workflows

Manager action	Assistant response
Configure clinic	Guided updates for clinic profile, contact information, doctors, schedules, working hours and fees.
Upload policy PDF/DOCX	Extracts text (with OCR for scans if needed), identifies rules, timings, fees and FAQ candidates.
Review and clarify	Creates a structured draft, asks about missing or contradictory information and waits for manager approval.
Activate approved information	Sends only approved changes to the backend; the AI never writes directly to the production database.

Language support	The AI recognises Hindi, English and common Hinglish phrasing; testing covers all three interaction styles.

03 / IMPLEMENTATION VIEW

Technology Stack

Each technology is retained only where it supports the Telegram-based MVP.

Technology	Responsibility in the MVP
Telegram Bot API	The only user-facing communication channel for patient and clinic-manager interactions.
Python	Runtime for the AI agent service and workflow orchestration.
FastAPI	Telegram webhook endpoint and agent API layer.
LangGraph	Coordinates multi-step workflows, state, approvals and escalation paths.
LangChain	LLM integration and carefully defined tools where useful.
Node.js + Express	Existing backend for authorization, clinic operations, appointment transactions and audit records.
Prisma ORM + PostgreSQL	Controlled access to clinic, doctor, patient, schedule and appointment records.
PDF/DOCX parser + OCR	Extracts clinic information from manager uploads; OCR is used for scanned PDFs when needed.
n8n (optional)	Non-critical notifications and background automation, not a dependency for the core MVP.

Design principle

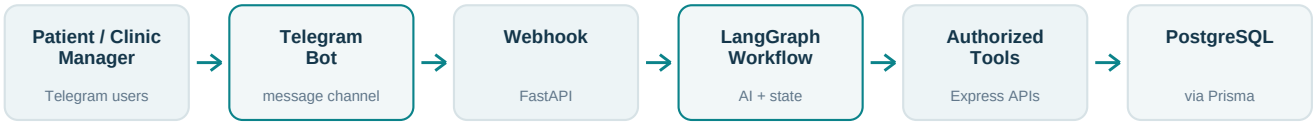
LangGraph manages the conversation and workflow state. Node.js remains the system of record for authorization, database operations, appointment transactions and audit trails. The agent can request an authorized backend action, but cannot directly alter production data.

04 / SYSTEM DESIGN

System Architecture and Workflow

A narrow, tool-mediated architecture keeps the AI assistant useful while preserving backend control.

MVP request path



Workflow coordination	Backend control
LangGraph decides the next permitted step: answer from approved data, ask for confirmation, request a tool call, ask a manager to clarify, or escalate to staff.	Node.js + Express validates roles, clinic boundaries and transaction rules before Prisma reads or writes PostgreSQL. Every approved action can be audited.

Mandatory control points

1	2	3
Role and clinic check before any protected operation.	Patient confirmation before booking, cancelling or rescheduling.	Manager approval before extracted policy information becomes active.

The architecture intentionally excludes PSTN/SIP, media gateways, audio streaming, realtime speech-to-speech and telephony vendors from the MVP.

05 / KEY FLOWS

Patient and Document Workflows

Both workflows preserve a clear point of confirmation before a consequential change takes effect.

Patient booking workflow

- Patient asks for a doctor or slot
- AI reads approved doctors and availability
- AI presents suitable slots
- Patient explicitly confirms
- Authorized backend creates appointment
- AI returns confirmation or escalates

Document ingestion workflow

- Manager sends PDF or DOCX in Telegram
- FastAPI extracts text; OCR if scan
- AI identifies policies, times, fees and FAQ candidates
- Draft is shown to manager
- Manager clarifies gaps and explicitly approves
- Backend saves approved configuration

Patient safeguard	Document safeguard
If information is unclear or verification fails, the workflow stops and clinic staff handle the exception. The assistant does not infer medical advice.	Extraction produces a **draft**, never a live policy. Missing or contradictory information is surfaced to the manager before approval.

Only manager-approved clinic information is eligible for patient-facing answers.

06 / DELIVERY OWNERSHIP

Team Structure and Task Distribution

Four members own clear workstreams and share integration quality responsibility.

Member	Primary ownership	Interfaces / deliverables
Aman Gupta Backend and Database	Node.js/Express, Prisma/PostgreSQL, clinic/doctor/patient/appointment APIs, availability/booking transactions, auth, authorization and clinic isolation.	Authorized APIs for the agent; validation, audit records and transaction test scenarios.
Pankaj Kumar AI Agent and Workflow	Python/FastAPI agent service, LangGraph workflow, LangChain/LLM integration, intent/context, Hindi/Hinglish understanding, tool coordination.	Tool schemas and workflow state shared with Aman, Abhinav and Nayan; escalation and approval rules.
Abhinav Kumar Telegram Bot and Integration	Bot creation/configuration, Bot API, webhooks, patient/manager interaction flows, message and document handling, Hindi/Hinglish tests.	Webhook contract to FastAPI; usable conversation flows and channel-level test cases.
Nayan Raj Document Processing, Testing and Integration	PDF/DOCX upload processing, extraction/OCR, policy/FAQ draft workflow, cross-service integration tests, end-to-end tests and documentation.	Document extraction interface; review/approval test cases and final integration evidence.

Shared collaboration model

Aman + Pankaj

Validate Telegram messages, documents and language behaviour against real conversation flows.

Nayan + Abhinav

Agree tool/API contracts and test authorization, confirmation and error paths together.

Whole team

Run end-to-end tests, review patient safety boundaries and keep the proposal/documentation current.

07 / OPERATING MODEL

Automation Level and Human Involvement

The aim is maximum automation for routine tasks, with deliberate review where information or consequences require it.

Stage	Automation	Human involvement
Patient FAQ	Automated	0 - answers use approved information only.
Doctor discovery	Automated	0
Availability checking	Automated	0
Appointment booking	Automated after confirmation	Patient explicitly confirms before the transaction.
Policy document extraction	Automated	0 initially; AI creates a draft.
Policy activation	AI prepares draft	Manager reviews, clarifies and approves.
Unclear requests	AI detects and escalates	Staff handles the exception.

Essential protections

Clinic data isolation	Role-based access	Minimal patient data
Backend controls ensure a clinic accesses only its own approved information and records.	Manager actions and protected patient operations are authorized by the backend, not by model output alone.	Telegram is not treated as automatically appropriate for sensitive medical records. Keep messages minimal and verify protected actions securely.

Human approval is required to activate extracted clinic policies; patients must confirm consequential appointment changes.

08 / BOUNDARY AND NEXT STEPS

MVP Scope and Future Enhancements

The proposal separates what will be demonstrated now from possibilities that would require a later project phase.

Included in Telegram MVP	Future possibilities - not part of this proposal
Telegram messages and document uploads Approved FAQ answers Doctor discovery and slot checking Confirmed booking, lookup, cancellation and rescheduling Clinic configuration and policy-draft approval Hindi, English and Hinglish interactions	Phone calling and telephony integrations Realtime speech-to-speech interaction Audio streaming, media gateways and barge-in PSTN/SIP infrastructure Broader production-scale SaaS operations

Conclusion

This MVP turns the broad AI healthcare receptionist concept into a practical college project: a Telegram-based assistant for routine clinic information and appointment workflows. Its value comes from approved information, controlled backend actions and clear human review - not from unnecessary voice infrastructure.

By keeping the communication channel narrow and the control points explicit, the team can demonstrate a credible end-to-end system while protecting clinic data, requiring confirmation for meaningful changes and avoiding medical advice.

PROJECT OUTCOME

A focused Telegram MVP that reduces routine reception work while keeping managers and staff in control of information and exceptions.